01 / EXECUTIVE INVESTMENT THESIS

Preserve readiness, gate irreversible capital

Decision. BHP should maintain engineering, permitting and execution readiness for the Escondida New Concentrator while staging major irreversible commitments ahead of its targeted calendar 2027-28 final investment decision. The recommendation preserves the replacement timetable and makes further commitments conditional on evidence that the project protects sufficient production under conservative long-term margins. Public disclosures support this decision framework, but do not establish an independently valued project or an optimal sanction date.

Two price horizons. Goldman Sachs' reported $15,400/t average forecast for calendar 2027 is relevant to near-term funding flexibility. First production at the new facility is targeted for 2031-32. BHP's disclosed 16-18% project IRR uses UBS long-term consensus copper of $4.76/lb as of May 2026, equivalent to about $10,494/t. Applying $15,400/t for the facility's operating life would change the investment thesis without a supporting source. The model therefore keeps near-term and long-term prices independent.

Replacement changes the investment question. The 230-270ktpa production range is average output from the new facility in FY2034-43. Los Colorados approaches the end of its operating life and the new concentrator replaces its production capacity. Economic benefit includes production that would otherwise disappear, plus actual improvements in recovery or throughput. Counting the full facility output as new mine growth would overstate additional supply. Excluding all replacement output from benefits would also miss the value protected by continuity.

What the screen establishes. At a $5.85bn gross capital midpoint, a 10% nominal discount rate, five equal year-end construction payments and 20 years of full benefits from 2032, capital recovery requires about $839m of annual after-tax benefit. A $4,000/t hypothetical all-in cash burden and 35% cash-tax wedge translate that into 199ktpa of effective preserved or incremental production. That is a hurdle to substantiate, not an estimate of the project's actual benefit or return.

Principal output	Result	Interpretation
GS revision sales exposure	$2.39bn	Annualised consolidated gross proxy
Base production-benefit hurdle	199ktpa	79.5% of 250ktpa facility midpoint
Combined stress hurdle	416ktpa	Exceeds disclosed facility output range

Independent SMPT research. BHP and Goldman Sachs are research subjects and have no sponsoring or advisory relationship with this publication.

Sources: **S01** **S05** **S08** **S09**

02 / COMMODITY MARKETS AND OPERATING EXPOSURE

A bullish bank forecast is already near realised prices

Forecast verification. The World Investment Institute reports a Goldman Sachs note dated 19 October 2026 raising the 2027 copper average from $13,800/t to $15,400/t and the end-2026 forecast to $14,700/t. International Copper Association Australia reproduces the same targets. An authorized original note was not located on the public Goldman Sachs hub. FinancialJuice exposed no substantive text, and the linked GBM coverage returned an access error. The publication retains explicit secondary attribution and does not reproduce proprietary bank material.

Competing explanations. Public coverage attributes the bank's view to US inventory accumulation, Chinese substitution from scrap toward refined copper, weaker mine supply and investor positioning. These mechanisms affect timing and location as well as aggregate demand. A deficit outside the US can coexist with a global surplus when metal accumulates in US inventories. ICAA reports an October ICSG global surplus outlook, but the primary October numerical release was not retrieved. It is used as an attributed competing interpretation, not an independently verified balance forecast.

A more conservative institutional reference. The World Bank's 27 October outlook forecasts $13,000/t for 2027 and $12,500/t for 2028. Goldman's reported 2027 figure is 18.5% higher. The World Bank supplies a useful alternative annual price case, not a market-forward curve or a prediction for the 2030s. The model's $8,500/t long-term downside and $13,000/t upside are independent stresses. Neither is presented as a newly surveyed consensus.

Operating evidence limits price extrapolation. BHP reported Q1 FY2027 copper production of 400.7kt, down 19%, and provisional realised copper of $7.01/lb, up 53%. The latter equals roughly $15,454/t, already close to Goldman's 2027 forecast. Multiplying the rounded price and volume indices gives about 24% higher gross value, but that is not reported revenue or EBITDA. Payable sales, final pricing, inventories, costs, taxes and by-products break the simple proportional relationship. Escondida's lower grades, maintenance, weather and safety suspension demonstrate why volume must remain an independent driver.

Copper price reference	US$/t	Horizon
GS revised, secondary reporting	15,400	2027 average
World Bank, primary forecast	13,000	2027 average
BHP IRR assumption via UBS	10,494	Long term
BHP Q1 realised, provisional	15,454	Jul-Sep 2026

Trade-policy effects enter the near-term price scenarios. No tariff recommendation is treated as enacted law, and no causal price attribution is independently estimated.

Sources: **S05** **S08** **S09** **S12**

03 / FINANCIAL PERFORMANCE AND OWNERSHIP

Copper exposure is material, but cash is not wholly attributable

Historical earnings and reporting boundaries. BHP's statutory Copper segment generated $29.031bn of FY2026 revenue and $18.187bn of underlying EBITDA, compared with $22.530bn and $12.326bn in FY2025. Group underlying EBITDA was $32.947bn. The published 54% copper contribution excludes Group and unallocated items from its denominator. A direct copper-to-total-group division gives 55.2%. The asset-management table includes a different equity-accounting presentation and must not be substituted into a statutory revenue comparison.

Consolidation differs from economic interest. BHP owns 57.5% of Escondida, Rio Tinto 30% and JECO interests a combined 12.5%. BHP controls and fully consolidates Escondida. Its $17.054bn FY2026 asset revenue and $12.440bn underlying EBITDA are gross consolidated figures. Multiplying every group metric by 57.5% would be incorrect. The annual report also identifies protective minority rights and restrictions on asset transfers and dividend payments. Pro rata economic exposure does not establish unrestricted access to mine cash.

A transparent annualised exposure bridge. The model uses FY2027 asset guidance midpoints of 1,050kt for Escondida, 220kt for Spence and 305kt for Copper South Australia: 1.575Mt on a consolidated production basis. An assumed 95% payable/sales factor gives a 1.496Mt sales proxy. A fixed-volume $1,600/t GS revision then produces $2.394bn of gross sales exposure. A 5% price-linked leakage assumption reduces the EBITDA proxy to $2.274bn. A 35% marginal cash-tax wedge reduces the cash proxy to $1.478bn before incremental minority distributions.

What the bridge omits. Antamina's equity-accounted production is kept separate, as are by-product effects. The sales factor, leakage, tax and $4,000/t avoidable cash cost used for volume stresses are SMPT assumptions, not company guidance. Different realised terms and provisional-pricing adjustments can alter timing and direction. Applying a calendar 2027 price to fiscal 2027 volume guidance provides a scale of exposure only. It cannot establish a forecast fiscal-year EBITDA, attributable profit or parent free cash flow.

US$bn unless labelled	FY2025	FY2026
Operating cash flow	18.692	21.778
Investing cash flow	(13.350)	(12.011)
Reported free cash flow	5.342	9.767
Net debt	12.924	8.694
Copper production (Mt, mixed basis)	2.017	1.953

Reported FCF already includes investing cash flows. The model never subtracts the same capex again from reported FCF.

Sources: **S02** **S03** **S05** **S06**

04 / ESCONDIDA INVESTMENT ECONOMICS

The project hurdle depends on production lost without replacement

Scope and estimate evolution. The March announcement indicated $4.4-5.9bn of capital and 220-260ktpa of facility output. August guidance increased these ranges to $5.4-6.3bn and 230-270ktpa and cited 10% higher concentrator throughput to 50Mtpa. The capex midpoint rose 13.6%, from $5.15bn to $5.85bn, against a 4.2% increase in copper output midpoint. This is a changed scope and estimate, not evidence by itself of a like-for-like construction overrun. BHP also raised its expected IRR by 2-3 percentage points. Public data cannot independently reconcile all contributors to that return improvement.

Approval and economic bases. August's approximately $0.5bn pre-commitment funding is explicitly BHP share. October reports $0.8bn on a 100% basis. These figures are not additive, and approval does not prove cash has been spent. Full FID remains targeted for CY2027-28. The $5.4-6.3bn total is a gross nominal estimate. Its 57.5% BHP economic share is $3.105-3.623bn. Other pipeline projects use real July 2026 dollars, so unadjusted portfolio totals would mix bases. Estimates also exclude specified commercial commitment investments.

Quantitative decision boundary. The base model discounts $5.85bn of assumed equal 2027-31 payments to $4.435bn at end-2026. Discounted full annual benefits from 2032 for 20 years have a factor of 5.286. Dividing capital by that factor gives the $839m annual after-tax hurdle. At $10,494/t less $4,000/t all-in burden and the 35% cash-tax wedge, the necessary benefit is 198.8ktpa. This assumes a flat full benefit before BHP's disclosed FY2034-43 averaging window, omits an engineering ramp-up and requires an independently chosen benefit life.

The information needed for sanction. A financing-grade analysis requires Los Colorados' no-build decline and closure profile, alternative ore-routing capacity, recovery and grade schedules, actual incremental and preserved volumes, annual capex draws, operating costs, fiscal depreciation, working capital, closure and by-product cash flows. Without them, the model cannot reproduce the 16-18% issuer IRR or estimate a reliable independent project NPV. It instead identifies how much economic continuity must be demonstrated and how that hurdle changes under stress.

Screen case	Required benefit ktpa	Share of 250ktpa
Base	199	80%
Upside	143	57%
Downside	287	115%
Delay	219	87%
Overrun	239	95%
Combined stress	416	167%

In the downside, 287ktpa exceeds the 270ktpa range ceiling under these copper-only assumptions. This can be remedied by better margins or other real benefits, not by relabelling gross facility output as incremental growth.

Sources: S01 S05 S07

05 / FUNDING CAPACITY AND STRATEGIC ALTERNATIVES

Capital staging protects flexibility only if continuity survives

Portfolio funding is more than a copper question. BHP guides approximately $11bn of annual capital and exploration expenditure through the medium term, including about $4bn of average annual growth spending. More than 55% of that growth is copper focused, rising to 66% on the stated basis including non-operated JV investment. Jansen Stage 2's total estimate increased from $4.9bn to $6.9bn. A $2.3bn impairment signals the cost of assuming construction budgets remain stable. Project budgets are totals across years and cannot be added to one annual group capex number.

Historical flexibility is substantial but qualified. At 30 June 2026, BHP reported $18.532bn of cash, including $87m restricted, and an undrawn $5.5bn revolving credit facility backing commercial paper. The $20bn upper net-debt target is a policy bound, not an undrawn bank commitment. FY2026's $4.3bn silver-streaming receipt is a financing liability cash flow and is excluded from recurring funding capacity. Dividends announced for FY2026 differ from cash paid during that year. These distinctions materially affect a cash-use analysis.

Cash-use illustration. The funding bridge holds FY2026 operating cash flow and other investing uses as anchors, substitutes $11bn group capex guidance and applies the modeled copper price-volume shock. It retains $6.756bn of BHP cash dividends and $2.355bn of NCI dividends, then adjusts the Escondida minority cash claim proportionately. The resulting annual residual is about $0.14bn in the base and a $1.65bn deficit in the downside. This mixed-period illustration highlights the need to flex distributions or financing under stress. It is not a forecast of future debt, and does not include a statutory debt maturity roll-forward.

Compare realistic options. Maintaining sequence retains the base 199ktpa hurdle. An illustrative one-year acceleration with a 10% execution premium raises it to 219ktpa, even with earlier benefits. A one-year shift of both capital and benefits with 5% escalation raises it to 209ktpa. Staging defers $1.17bn of gross first-year expenditure, about $0.67bn pro rata to BHP, but does not eliminate the investment. At an assumed 125ktpa of production at risk, a lost year represents roughly $528m of after-tax benefit at base margins. Actual loss depends on the counterfactual and available processing alternatives.

Alternative	Priority	Required evidence
Maintain	Protect replacement schedule	Benefit clears downside hurdle
Accelerate selected work	Prevent avoidable continuity loss	Permits, safe capacity, avoided loss > premium
Stage major awards	Retain financing and scope choices	Deferral leaves sufficient continuity

The recommendation favors staged decision gates within a maintained readiness program. It does not assume that a full one-year project delay is optimal.

Sources: **S01** **S02** **S03**

06 / RISKS, TRIGGERS AND PUBLICATION DECISION

Make the next commitment conditional on verifiable operating value

Risk transmission. US stockbuilding, scrap substitution and speculative positioning enter the model through near-term price. A slower US inventory build can ease ex-US scarcity even if industrial demand remains intact. Neither public forecast proves the policy mechanism caused a specific observed price move. Chilean permit timing, community process, water and power availability, labour and safe execution affect schedule and cost. Lower grades and recovery shortfalls affect effective production benefit. Currency, freight, energy and construction inflation can increase the assumed all-in burden or capex. No unsupported probability is assigned to these risks.

Management decision triggers. Before major awards, obtain an approved permitting path and execution-ready estimate, identify the annual production lost without the replacement, and rerun a complete after-tax project model including ramp-up and commercial commitments. Preserve the timetable if continuity value clears conservative cost and schedule cases. Accelerate only work that can be executed safely and whose avoided production loss exceeds the incremental premium. Re-scope or defer discretionary commitments when the conservative benefit remains below the hurdle and continuity can be protected through alternatives.

Balance-sheet and shareholder triggers. Review all-in portfolio cash needs against updated commodity and operating conditions, debt maturities and the minimum payout framework. The illustrative downside deficit supports testing distribution flexibility and sequencing, not an automatic conclusion that BHP cannot finance the project. A durable improvement in long-term margins or a larger verified no-build production loss would strengthen investment. Material overrun, delayed approvals or a smaller preserved benefit would weaken it. Current high prices alone do not resolve those questions.

Research conclusion and professional use. The analysis establishes a reproducible distinction between cash available to invest and the economic value created or protected by investing. It tests price and volume together, reconciles consolidated and economically attributable metrics, and identifies an observable capital-recovery hurdle. It is useful for corporate strategy screening, mining equity scenario discussions and investment-committee preparation. It is insufficient for credit underwriting, a definitive sanction decision or a trading recommendation.

Editorial gate and next capability. The package is READY FOR EDITORIAL REVIEW as a conditional strategy study. Public release requires a dated refresh of market and permit evidence, acceptance of the explicit secondary attribution for the GS forecast or authorized original verification, and approval of the investment-screen framing. Independent project valuation remains unavailable until the full technical and fiscal cash-flow inputs are obtained. The next highest-value capability is a reusable replacement-versus-expansion counterfactual schedule, applied to forthcoming BHP Copper South Australia disclosures when technical inputs are available.

Outstanding input	Why it matters
Los Colorados no-build decline	Determines preserved production benefit
Annual costs, tax and ramp-up	Determines actual project cash flows
Permit decision and execution plan	Constrains credible acceleration
Original GS report / dated refresh	Improves attribution and temporal reliability

Research completion does not confer publication authorization. No live-site changes, public distribution, deployment or messages to third parties are part of this assignment.

Sources: **S01** **S03** **S05** **S08** **S12**